MANU 465 Group Project Proposal

Preprocessing

Import Libraries

In [26]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

Import Datasets

In [27]:
```
data_test_FD1 = pd.read_csv('CMAPSSData/test_FD001.txt', sep=r"\s+", header=None)
data_train_FD1 = pd.read_csv('CMAPSSData/train_FD001.txt', sep=r"\s+", header=None)

X = data_train_FD1.iloc[:, 2:-1].values # all columns except last
y = data_train_FD1.iloc[:, 0].values # rows of last column only
```

Information on DataSet

Dataset Sample

In [28]:
```
data_train_FD1.sample(n=10, random_state=1)
```

Out[28]:

	0	1	2	3	4	5	6	7	8	9	...	16
8296	43	59	-0.0008	-0.0005	100.0	518.67	642.52	1579.68	1404.68	14.62	...	521.88
10031	51	123	0.0001	-0.0002	100.0	518.67	642.46	1592.13	1404.88	14.62	...	522.15
14957	75	28	0.0020	-0.0002	100.0	518.67	642.23	1585.87	1408.75	14.62	...	521.75
82	1	83	0.0004	0.0005	100.0	518.67	642.66	1588.05	1406.38	14.62	...	521.93
18428	91	49	-0.0009	-0.0004	100.0	518.67	642.37	1589.87	1416.56	14.62	...	521.77
11259	57	5	-0.0022	0.0004	100.0	518.67	642.48	1589.61	1408.72	14.62	...	521.80
11211	56	232	0.0016	0.0002	100.0	518.67	643.02	1594.17	1418.31	14.62	...	520.38
10592	54	63	0.0005	-0.0002	100.0	518.67	642.31	1576.09	1399.97	14.62	...	522.17
9883	50	173	-0.0007	0.0002	100.0	518.67	643.40	1597.41	1420.73	14.62	...	520.77
3354	17	50	-0.0002	-0.0002	100.0	518.67	642.03	1582.81	1399.94	14.62	...	522.68

10 rows × 26 columns

General Information

In [29]:
```
data_train_FD1.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20631 entries, 0 to 20630
Data columns (total 26 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   0       20631 non-null  int64
 1   1       20631 non-null  int64
 2   2       20631 non-null  float64
 3   3       20631 non-null  float64
 4   4       20631 non-null  float64
 5   5       20631 non-null  float64
 6   6       20631 non-null  float64
 7   7       20631 non-null  float64
 8   8       20631 non-null  float64
 9   9       20631 non-null  float64
 10  10      20631 non-null  float64
 11  11      20631 non-null  float64
 12  12      20631 non-null  float64
 13  13      20631 non-null  float64
 14  14      20631 non-null  float64
 15  15      20631 non-null  float64
 16  16      20631 non-null  float64
 17  17      20631 non-null  float64
 18  18      20631 non-null  float64
 19  19      20631 non-null  float64
 20  20      20631 non-null  float64
 21  21      20631 non-null  int64
 22  22      20631 non-null  int64
 23  23      20631 non-null  float64
 24  24      20631 non-null  float64
 25  25      20631 non-null  float64
dtypes: float64(22), int64(4)
memory usage: 4.1 MB
```

In [30]:
```
data_train_FD1.describe()
```

Out[30]:

	0	1	2	3	4	5	
count	20631.000000	20631.000000	20631.000000	20631.000000	20631.0	2.063100e+04	206
mean	51.506568	108.807862	-0.000009	0.000002	100.0	5.186700e+02	6
std	29.227633	68.880990	0.002187	0.000293	0.0	6.537152e-11	
min	1.000000	1.000000	-0.008700	-0.000600	100.0	5.186700e+02	6
25%	26.000000	52.000000	-0.001500	-0.000200	100.0	5.186700e+02	6
50%	52.000000	104.000000	0.000000	0.000000	100.0	5.186700e+02	6
75%	77.000000	156.000000	0.001500	0.000300	100.0	5.186700e+02	6
max	100.000000	362.000000	0.008700	0.000600	100.0	5.186700e+02	6

8 rows × 26 columns

Dataset Visualization

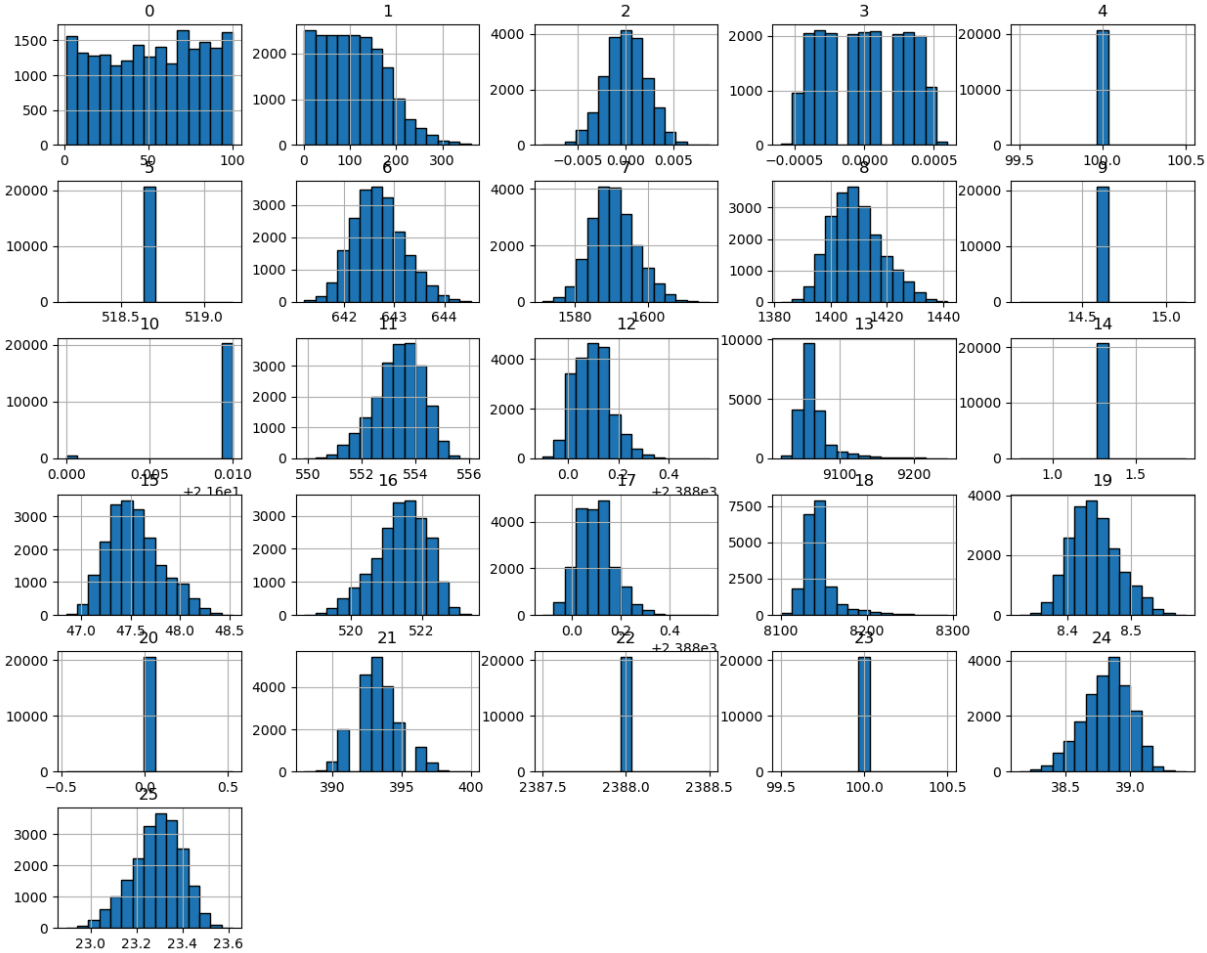
Histograms

In [31]:

```
data_train_FD1 = data_train_FD1.apply(pd.to_numeric, errors="coerce")

# Plot histograms
data_train_FD1.hist(figsize=(15, 12), bins=15, edgecolor='black')

# Show the plot
plt.show()
```


0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
+2.16e1
16
17
+2.388e3
18
19
20
21
22
+2.388e3
23
24
25